

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear inequalities in one or two variables	Hard

Question

A team hosting an event to raise money for new uniforms plans to sell at least **140** tickets before this event and at least **220** tickets during this event to raise a total of at least **\$5,820** from all tickets sold. The price of a ticket during this event is **\$3** less than the price of a ticket before this event. Which inequality represents this situation, where x is the price, in dollars, of a ticket sold during this event?

Answer

- A. $140(x + 3) + 220x \leq 5,820$
- B. $140(x + 3) + 220x \geq 5,820$
- C. $140(x - 3) + 220x \leq 5,820$
- D. $140(x - 3) + 220x \geq 5,820$

Correct Answer: B

Rationale

Choice B is correct. It's given that a team plans to sell at least **140** tickets before an event and at least **220** tickets during the event to raise a total of at least **\$5,820** from all tickets sold. It's also given that the price of a ticket during the event is **\$3** less than the price of a ticket before the event and that x represents the price, in dollars, of a ticket sold during the event. It follows that $x + 3$ represents the price, in dollars, of a ticket sold before the event. The expression $140(x + 3)$ represents the planned revenue, in dollars, from the tickets sold before the event, and the expression $220x$ represents the planned revenue, in dollars, from the tickets sold during the event. Thus, the expression $140(x + 3) + 220x$ represents the planned revenue, in dollars, from all tickets sold. Since the team plans to raise a total of at least **\$5,820** from all tickets sold, the total revenue must be at least **\$5,820**. Therefore, the inequality $140(x + 3) + 220x \geq 5,820$ represents this situation.

Choice A is incorrect. This inequality represents a situation in which the team raises a total of at most **\$5,820** from all tickets sold.

Choice C is incorrect. This inequality represents a situation in which the price of a ticket before the event is **\$3** less, rather than **\$3** more, than the price of a ticket during the event and the team raises a total of at most **\$5,820** from all tickets sold.

Choice D is incorrect. This inequality represents a situation in which the price of a ticket before the event is **\$3** less, rather than **\$3** more, than the price of a ticket during the event.